

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

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Time on Task:

0 hours, 38 minutes

Progress:

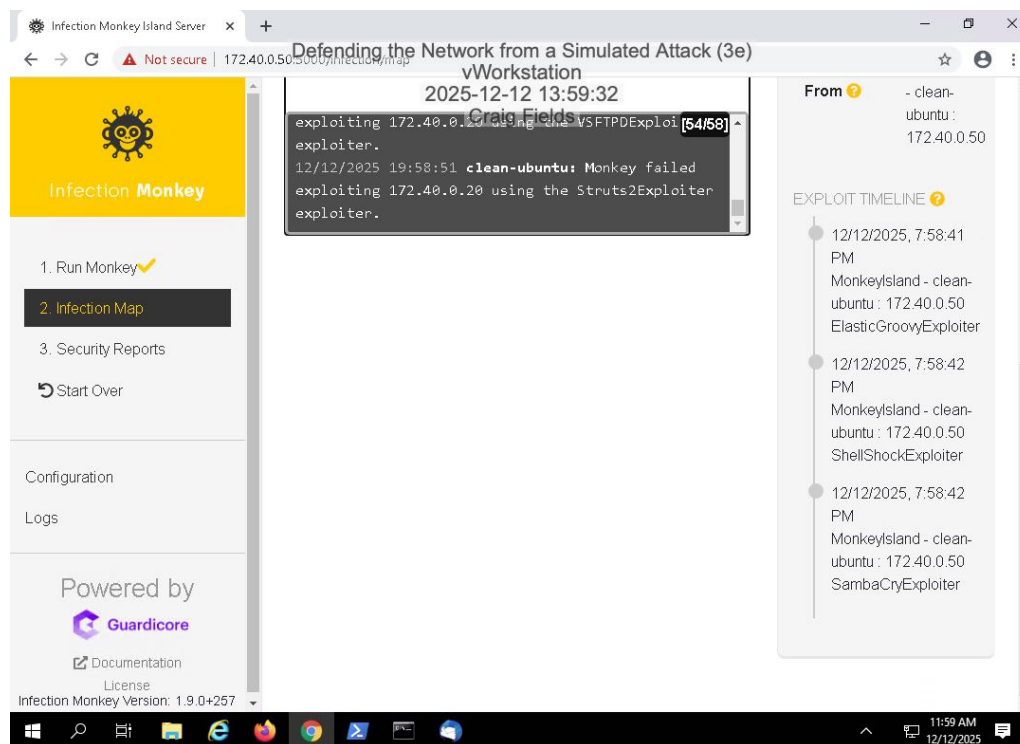
100%

Report Generated: Friday, December 12, 2025 at 3:30 PM

Section 1: Hands-On Demonstration

Part 1: Perform a Simulated Attack with Infection Monkey

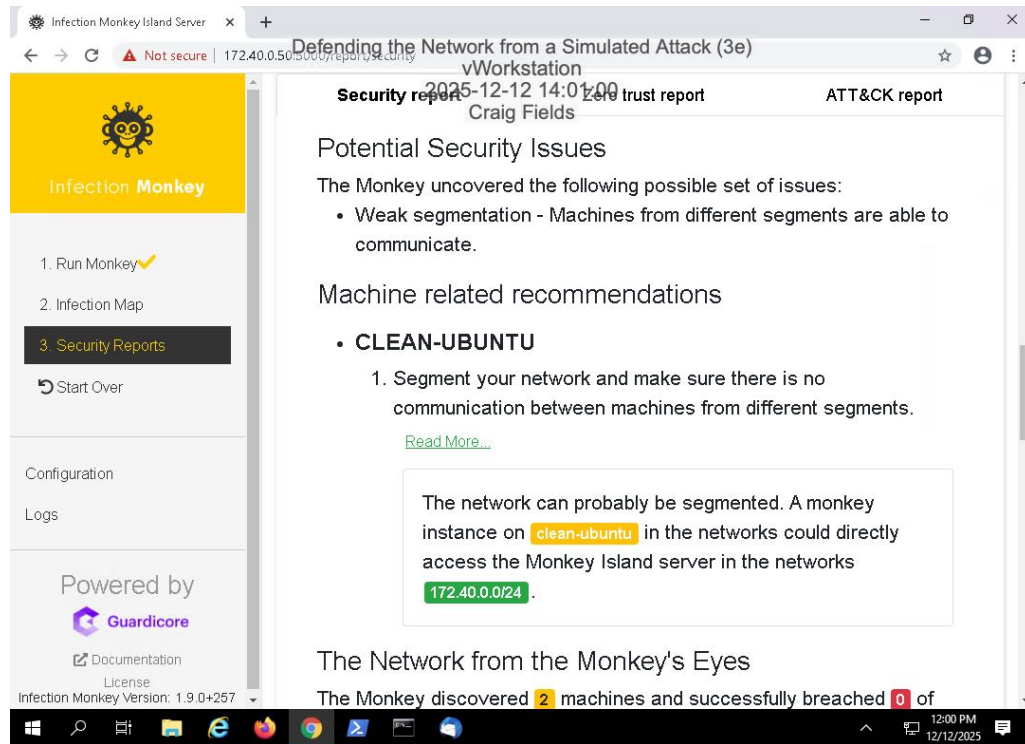
14. Make a screen capture showing the **successful exploit of the corporationtechs.com web server from MonkeyIsland.**



Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

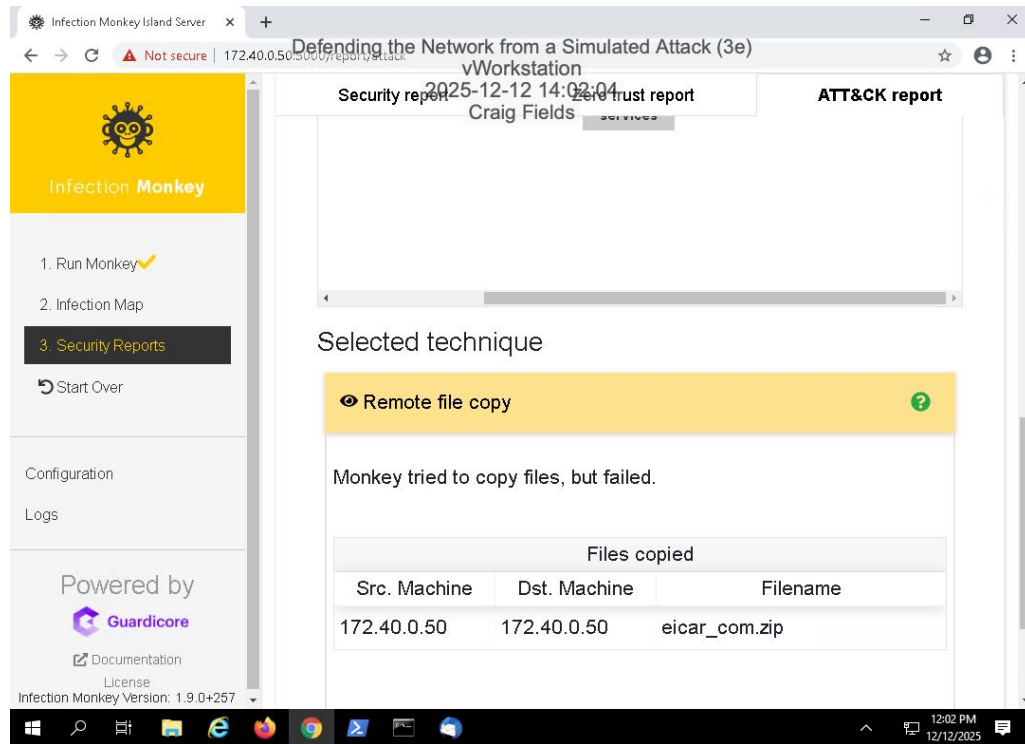
17. Make a screen capture showing the recommendations for the corporationtechs.com web server.



Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

20. **Make a screen capture** showing the **remote zip file** copied to the **corporationtechs.com** machine (172.40.0.20).

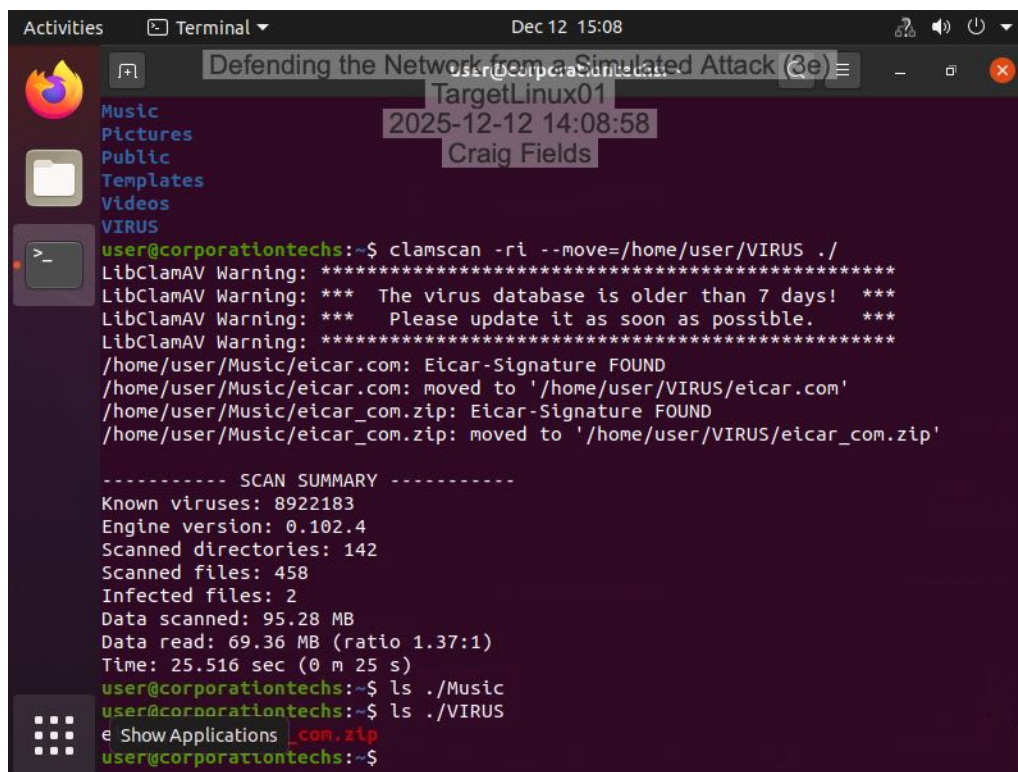


Part 2: Use Antivirus Software to Remove Malicious Files

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

12. Make a screen capture showing the contents of the VIRUS directory.



```
Activities Terminal Dec 12 15:08
Defending the Network from a Simulated Attack (3e)
TargetLinux01
2025-12-12 14:08:58
Craig Fields

Music
Pictures
Public
Templates
Videos
VIRUS
>_

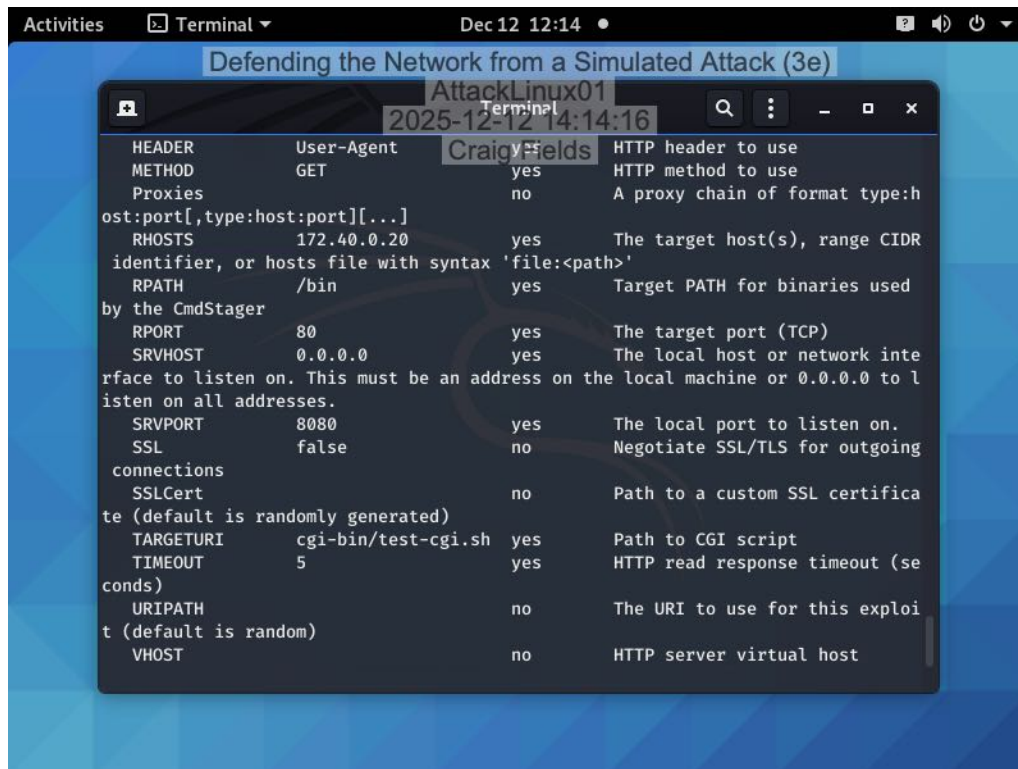
user@corporationtechs:~$ clamscan -ri --move=/home/user/VIRUS ./
LibClamAV Warning: *****
LibClamAV Warning: *** The virus database is older than 7 days! ***
LibClamAV Warning: *** Please update it as soon as possible. ***
LibClamAV Warning: *****
/home/user/Music/eicar.com: Eicar-Signature FOUND
/home/user/Music/eicar.com: moved to '/home/user/VIRUS/eicar.com'
/home/user/Music/eicar_com.zip: Eicar-Signature FOUND
/home/user/Music/eicar_com.zip: moved to '/home/user/VIRUS/eicar_com.zip'

----- SCAN SUMMARY -----
Known viruses: 8922183
Engine version: 0.102.4
Scanned directories: 142
Scanned files: 458
Infected files: 2
Data scanned: 95.28 MB
Data read: 69.36 MB (ratio 1.37:1)
Time: 25.516 sec (0 m 25 s)
user@corporationtechs:~$ ls ./Music
user@corporationtechs:~$ ls ./VIRUS
e Show Applications eicom.zip
user@corporationtechs:~$
```

Section 2: Applied Learning

Part 1: Exploit a Vulnerable Web Server with Metasploit

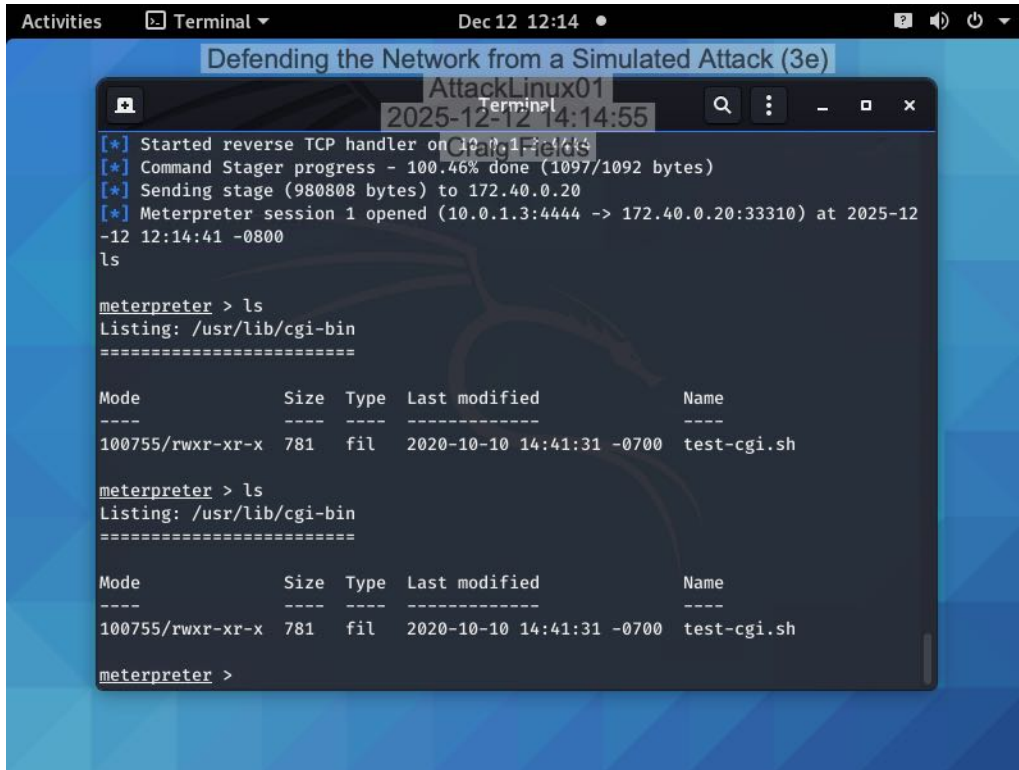
11. Make a screen capture showing the **updated exploit settings**.



Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

17. Make a screen capture showing the **successful Linux shell command** on TargetLinux01.



The screenshot shows a terminal window titled "Defending the Network from a Simulated Attack (3e)" with a subtitle "AttackLinux01". The terminal output indicates a successful reverse TCP handler connection and a Meterpreter session opening. The user runs the 'ls' command, listing the contents of /usr/lib/cgi-bin, which shows a file named 'test-cgi.sh' with permissions 100755/rwxr-xr-x and a size of 781 bytes.

```
2025-12-12 14:14:55
[*] Started reverse TCP handler on 10.0.1.3:4444
[*] Command Stager progress - 100.46% done (1097/1092 bytes)
[*] Sending stage (980808 bytes) to 172.40.0.20
[*] Meterpreter session 1 opened (10.0.1.3:4444 -> 172.40.0.20:33310) at 2025-12-12 12:14:41 -0800
ls
meterpreter > ls
Listing: /usr/lib/cgi-bin
=====
Mode                Size  Type  Last modified             Name
----                -
100755/rwxr-xr-x    781  fil   2020-10-10 14:41:31 -0700 test-cgi.sh

meterpreter > ls
Listing: /usr/lib/cgi-bin
=====
Mode                Size  Type  Last modified             Name
----                -
100755/rwxr-xr-x    781  fil   2020-10-10 14:41:31 -0700 test-cgi.sh

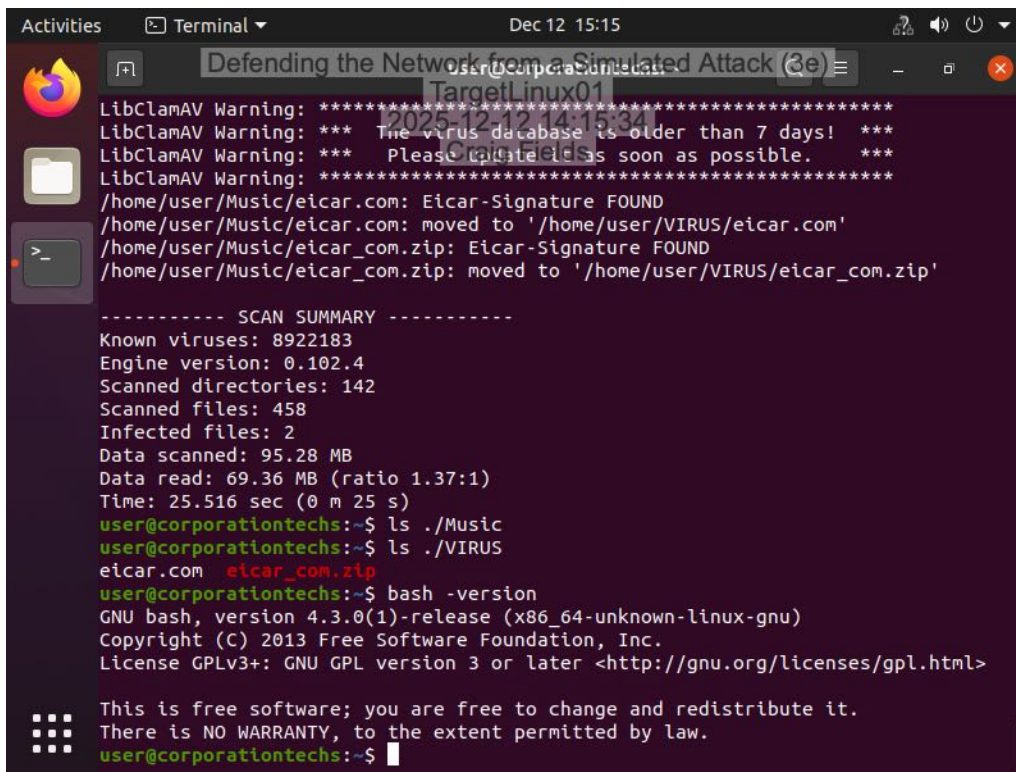
meterpreter >
```

Part 2: Patch the Exploited System

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

4. Make a screen capture showing the **pre-patch Bash** version.



The image shows a terminal window titled "Terminal" with a timestamp of "Dec 12 15:15". The window displays the output of a ClamAV scan. The scan results show two files, "eicar.com" and "eicar_com.zip", both identified as "Eicar-Signature FOUND". The files were moved to a directory named "/home/user/VIRUS/". Below the scan results, a "SCAN SUMMARY" is displayed, showing statistics such as "Known viruses: 8922183", "Engine version: 0.102.4", and "Scanned files: 458". The terminal also shows the output of the "bash -version" command, which displays the GNU bash version 4.3.0(1)-release (x86_64-unknown-linux-gnu) and its copyright information.

```
Activities  Terminal  Dec 12 15:15
Defending the Network from a Simulated Attack (3e)
Target: linux01
2025-12-12 14:15:34
ClamAV
LibClamAV Warning: *****
LibClamAV Warning: *** The virus database is older than 7 days! ***
LibClamAV Warning: *** Please update it as soon as possible. ***
LibClamAV Warning: *****
/home/user/Music/eicar.com: Eicar-Signature FOUND
/home/user/Music/eicar.com: moved to '/home/user/VIRUS/eicar.com'
/home/user/Music/eicar_com.zip: Eicar-Signature FOUND
/home/user/Music/eicar_com.zip: moved to '/home/user/VIRUS/eicar_com.zip'

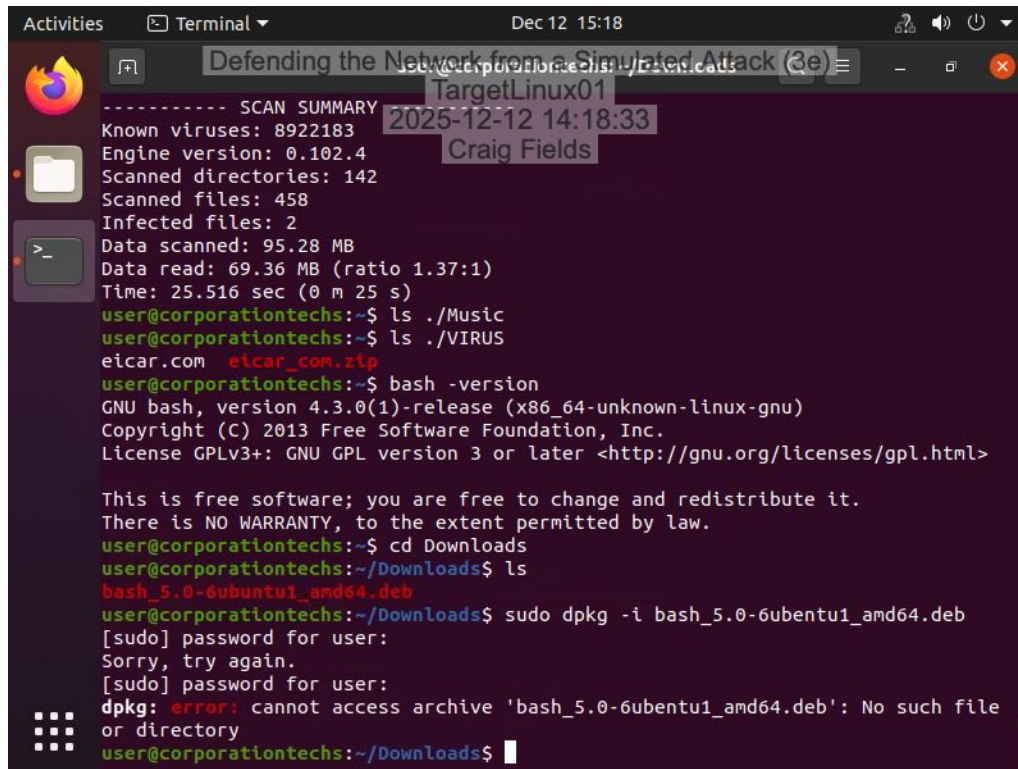
----- SCAN SUMMARY -----
Known viruses: 8922183
Engine version: 0.102.4
Scanned directories: 142
Scanned files: 458
Infected files: 2
Data scanned: 95.28 MB
Data read: 69.36 MB (ratio 1.37:1)
Time: 25.516 sec (0 m 25 s)
user@corporationtechs:~$ ls ./Music
user@corporationtechs:~$ ls ./VIRUS
eicar.com  eicar_com.zip
user@corporationtechs:~$ bash -version
GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$
```


Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

9. Make a screen capture showing the **post-patch Bash** version.

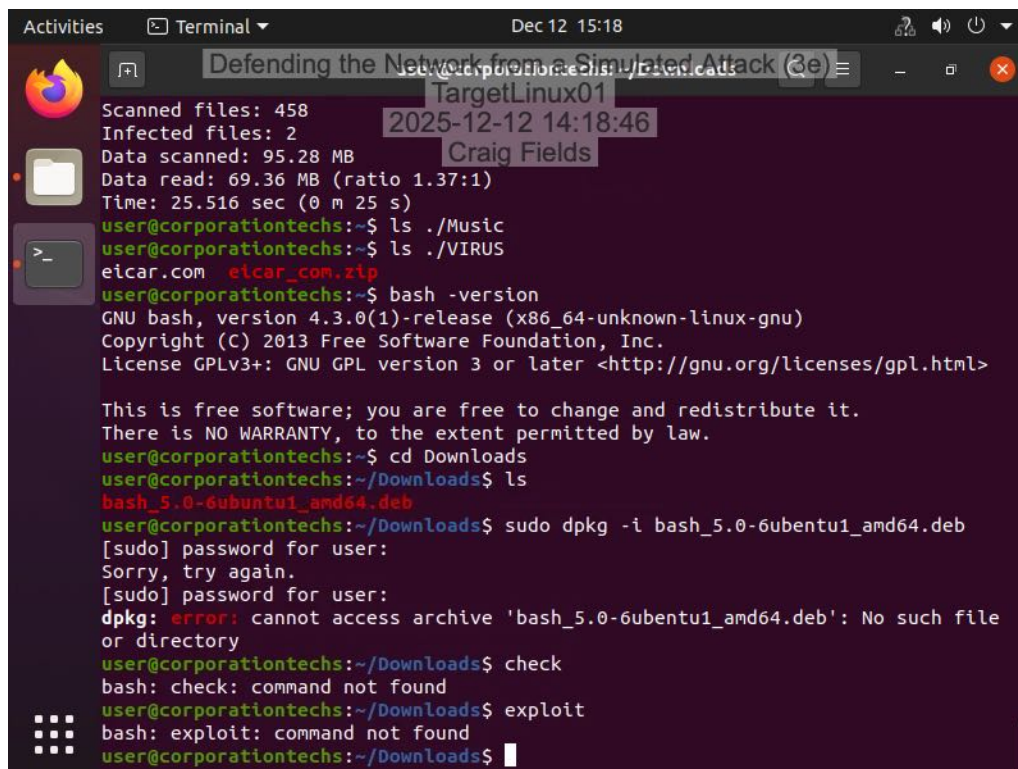


A terminal window titled "Defending the Network from a Simulated Attack (3e)" with a subtitle "TargetLinux01". The terminal shows a scan summary with the following details: Known viruses: 8922183, Engine version: 0.102.4, Scanned directories: 142, Scanned files: 458, Infected files: 2, Data scanned: 95.28 MB, Data read: 69.36 MB (ratio 1.37:1), Time: 25.516 sec (0 m 25 s). The user then runs 'ls ./Music' and 'ls ./VIRUS', showing 'eicar.com' and 'eicar_com.zip'. Next, they run 'bash -version', displaying 'GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)' and copyright information. They then run 'cd Downloads' and 'ls', showing 'bash_5.0-6ubuntu1_amd64.deb'. Finally, they attempt to install it with 'sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb', but receive an error: 'dpkg: error: cannot access archive 'bash_5.0-6ubuntu1_amd64.deb': No such file or directory'.

```
----- SCAN SUMMARY
Known viruses: 8922183
Engine version: 0.102.4
Scanned directories: 142
Scanned files: 458
Infected files: 2
Data scanned: 95.28 MB
Data read: 69.36 MB (ratio 1.37:1)
Time: 25.516 sec (0 m 25 s)
user@corporationtechs:~$ ls ./Music
user@corporationtechs:~$ ls ./VIRUS
eicar.com  eicar_com.zip
user@corporationtechs:~$ bash -version
GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$ cd Downloads
user@corporationtechs:~/Downloads$ ls
bash_5.0-6ubuntu1_amd64.deb
user@corporationtechs:~/Downloads$ sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb
[sudo] password for user:
Sorry, try again.
[sudo] password for user:
dpkg: error: cannot access archive 'bash_5.0-6ubuntu1_amd64.deb': No such file
or directory
user@corporationtechs:~/Downloads$
```

13. Make a screen capture showing your **unsuccessful exploit attempt**.



A terminal window titled "Defending the Network from a Simulated Attack (3e)" with a subtitle "TargetLinux01". The terminal shows the same scan summary as the previous screenshot. The user then runs 'bash -version' and 'cd Downloads'. They attempt to install 'bash_5.0-6ubuntu1_amd64.deb' with 'sudo dpkg -i', but receive the same error: 'dpkg: error: cannot access archive 'bash_5.0-6ubuntu1_amd64.deb': No such file or directory'. They then run 'check', 'exploit', and 'bash: check: command not found', 'bash: exploit: command not found'.

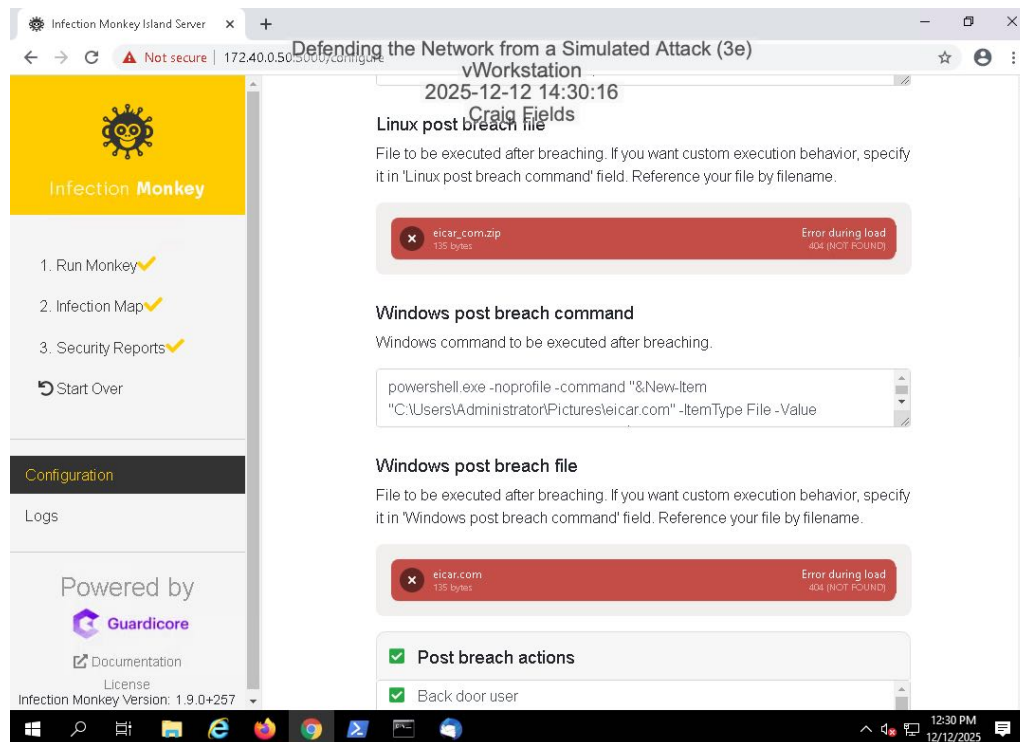
```
Scanned files: 458
Infected files: 2
Data scanned: 95.28 MB
Data read: 69.36 MB (ratio 1.37:1)
Time: 25.516 sec (0 m 25 s)
user@corporationtechs:~$ ls ./Music
user@corporationtechs:~$ ls ./VIRUS
eicar.com  eicar_com.zip
user@corporationtechs:~$ bash -version
GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$ cd Downloads
user@corporationtechs:~/Downloads$ ls
bash_5.0-6ubuntu1_amd64.deb
user@corporationtechs:~/Downloads$ sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb
[sudo] password for user:
Sorry, try again.
[sudo] password for user:
dpkg: error: cannot access archive 'bash_5.0-6ubuntu1_amd64.deb': No such file
or directory
user@corporationtechs:~/Downloads$ check
bash: check: command not found
user@corporationtechs:~/Downloads$ exploit
bash: exploit: command not found
user@corporationtechs:~/Downloads$
```


Section 3: Challenge and Analysis

Part 1: Run an Antivirus Scan on the vWorkstation

Make a screen capture showing the **EICAR file discovered by Windows Virus and threat protection.**



Part 2: Harden the Network Perimeter

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

Make a screen capture showing the **updated** firewall rules on the DMZ interface.

The screenshot shows the pfSense web interface in a browser window. The address bar displays the URL `172.30.0.1`. The page title is "Defending the Network from a Simulated Attack (3e)". The interface includes a navigation menu with options like System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. A warning message at the top states: "WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager." The main content area is divided into two columns. The left column, titled "System Information", contains details about the system, including the name, user, system type, BIOS, version, and CPU type. The right column, titled "Interfaces", shows a table of network interfaces. Below the interfaces table is a section for "Snort Alerts".

System Information			
Name	pfSense.securelabsondemand.com		
User	admin@172.30.0.2 (Local Database)		
System	VMware Virtual Machine Netgate Device ID: f5547cfed5be4caeacf		
BIOS	Vendor: Phoenix Technologies LTD Version: 6.00 Release Date: Wed Dec 12 2018		
Version	2.4.5-RELEASE (amd64) built on Tue Mar 24 15:25:50 EDT 2020 FreeBSD 11.3-STABLE Obtaining update status		
CPU Type	Intel(R) Xeon(R) Gold 6242 CPU @ 2.80GHz AES-NI CPU Crypto: Yes (inactive)		
Kernel BTI	Disabled		

Interfaces			
WAN	↑	autoselect	202.20.1.1
LAN	↑	autoselect	172.30.0.1
DMZ	↑	autoselect	172.40.0.1

Snort Alerts		
Interface/Time	Src/Dst Address	Description